

CAREER MENTOR

Project Report

An AI-powered multimodal interview preparation and evaluation platform

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Project Type	Full-stack AI interview simulator with CV, NLP, speech, analytics, and cloud storage
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1. Executive Summary

Career Mentor is an AI-powered interview preparation and evaluation platform designed to simulate real technical interview experiences. It combines resume analysis, personalized question generation, voice and coding interview rounds, webcam-based behavioral monitoring, AI evaluation, and automated PDF reporting into a single system.

The project evolved from a basic interview simulator into a multimodal assessment platform that brings together generative AI, speech processing, computer vision, cloud storage, and deployment automation. Its final form demonstrates end-to-end engineering thinking rather than just isolated model usage.

2. Why This Project Was Built

- To help students and job seekers practice interviews in a realistic, structured environment.
- To automate evaluation using Generative AI instead of relying only on manual self-assessment.
- To provide instant, detailed feedback with actionable improvement suggestions.
- To combine voice interviews, coding interviews, and behavioral analysis inside one unified platform.
- To create an AI-based end-to-end interview ecosystem similar to modern hiring workflows.

3. Problem Statement: What It Solves

- Lack of personalized mock interviews based on a candidate's own resume.
- No single practice platform that evaluates spoken answers, coding answers, and interview behavior together.
- Difficulty getting objective feedback on confidence, eye contact, posture, gesture movement, and cheating indicators.
- Slow and inconsistent manual review of responses and interview performance.
- No automatic generation of a polished performance report that can be saved, shared, or reviewed later.

4. Project Evolution: From Initial Idea to Final Product

Initially, Career Mentor was planned as a simple AI interview web application with resume upload and AI-generated questions. The first design focused on a small scope: resume upload, basic voice recording, simple answer evaluation, static PDF report generation, and Flask-React integration.

- Resume upload and AI question generation
- Basic voice recording
- Simple answer evaluation
- Static PDF report generation
- Frontend-backend integration using Flask and React

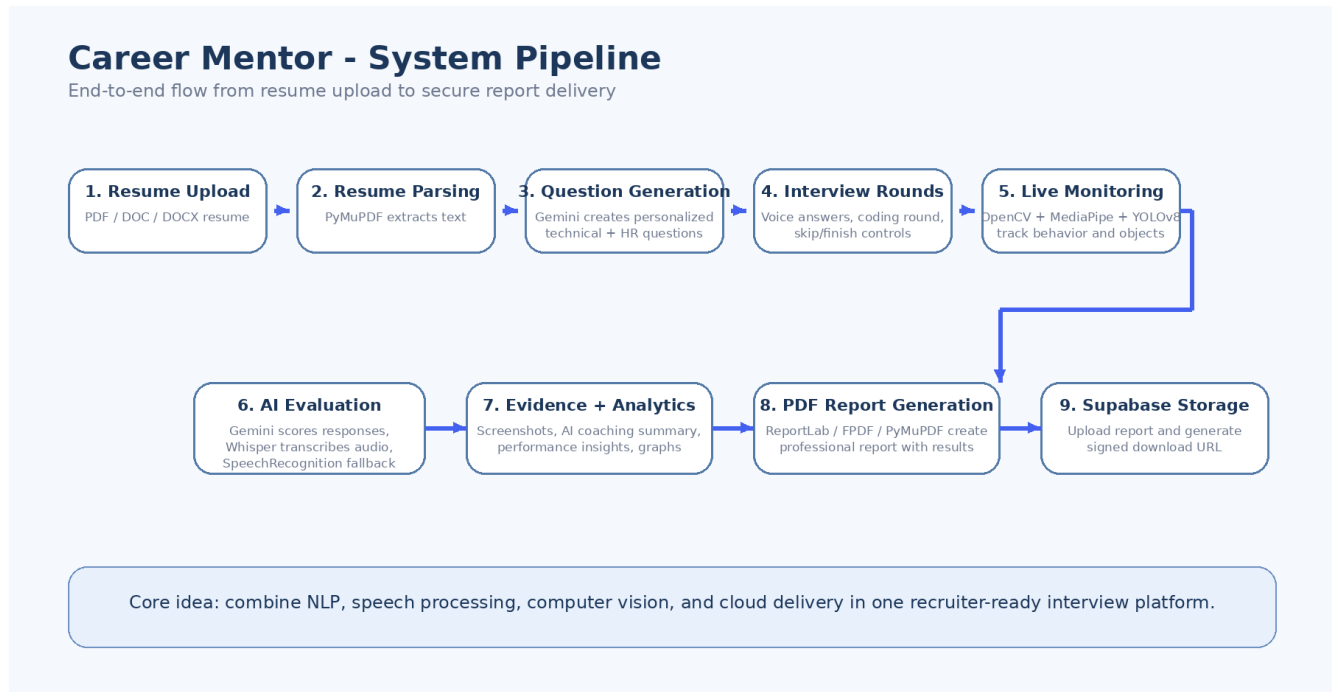
During development, the scope expanded significantly. The final platform added coding rounds, richer evaluation, live camera monitoring, analytics, cloud storage, and a more production-ready workflow.

- Monaco Editor for coding interview rounds
- Judge0 API for online code compilation and execution
- Whisper speech-to-text transcription
- Gemini-based detailed evaluation and scoring
- Live camera monitoring with OpenCV
- Supabase cloud storage
- Text-to-speech question narration
- Skip question and finish interview controls

- Detailed AI-generated interview reports with explanations

5. End-to-End Workflow and System Pipeline

The pipeline below shows the complete execution path of the system. It begins with resume upload and continues through AI question generation, interview interaction, behavioral monitoring, answer analysis, report creation, and secure storage.



In practical terms, the frontend handles the user-facing interview experience, while the backend orchestrates AI calls, transcription, monitoring, scoring, and report delivery.

6. Module-by-Module Architecture

- Frontend: handles UI, interview screens, code editor, animations, and analytics display.
- Backend: accepts requests, manages interview logic, coordinates AI services, and returns structured results.
- AI and NLP layer: generates questions, transcribes speech, scores answers, and writes coaching feedback.
- Computer vision layer: tracks eye contact, face landmarks, posture, hand movement, and suspicious objects.
- Report layer: converts raw scores and observations into a polished PDF report.
- Cloud layer: stores generated reports and exposes them through signed URLs.

7. Complete Technology Stack and Where Each Part Is Used

Technology	Role in Project	Section / Place Used
React.js	Frontend UI, interview workflow screens, state-driven user interaction	Frontend
TypeScript	Frontend type safety, cleaner component contracts, maintainability	Frontend
Vite	Fast frontend bundling and development server	Frontend
Tailwind CSS	Responsive styling and modern visual layout	Frontend

shadcn/ui	Reusable UI components for professional interface elements	Frontend
Framer Motion	Smooth animations and transitions in the frontend	Frontend
Monaco Editor	Browser-based coding environment for programming rounds	Frontend
Recharts	Analytics graphs and performance visualization	Frontend
Flask	Lightweight backend API and AI orchestration layer	Backend / orchestration
Flask-CORS	Frontend-backend communication across origins	Backend / orchestration
Gunicorn	Production WSGI server for deployment on Render	Backend / orchestration
Google Gemini API	Question generation, answer evaluation, coaching feedback, and final assessment	AI evaluation & transcription
OpenAI Whisper	Speech-to-text transcription of spoken answers	AI evaluation & transcription
SpeechRecognition	Fallback transcription when Whisper fails or is unavailable	AI evaluation & transcription
OpenCV	Camera access, frame capture, frame processing, and image handling	Live monitoring / computer vision
MediaPipe	Face landmarks, pose estimation, hand tracking, eye-contact and posture analysis	Live monitoring / computer vision
YOLOv8	Object detection and cheating/suspicious-object detection	Live monitoring / computer vision
Judge0 API	Code execution and compilation service for coding rounds	Coding round
RapidAPI	Gateway used to access Judge0 endpoints	Coding round
Supabase Storage	Cloud storage for reports, resumes, and secure downloads	Cloud storage / persistence
Supabase PostgreSQL	Planned session persistence and user data management	Cloud storage / persistence
ReportLab	Structured PDF report generation	Reports / parsing
FPDF	Additional PDF generation support	Reports / parsing
PyMuPDF	Resume parsing and PDF text extraction	Reports / parsing
Docker	Containerization for stable deployment and dependency isolation	Deployment / DevOps
Render	Hosting and deployment platform for frontend and backend	Deployment / DevOps
GitHub	Version control and repository management	Deployment / DevOps
Threading	Asynchronous monitoring so long-running tasks do not block requests	Backend / orchestration
UUID	Unique session identifiers for interview runs	Backend / orchestration
JSON processing	Parsing and validating AI responses and backend payloads	Backend / orchestration

8. AI Models and Intelligent Components

Model / Component	Task	Why It Was Used	Where It Appears
Gemini	Generate questions, score answers, create feedback	Strong LLM support for interview-style reasoning and structured evaluation	Question generation, answer evaluation, final report
Whisper	Speech-to-text transcription	Reliable transcription for spoken interview answers	Voice interview analysis pipeline
SpeechRecognition	Fallback transcription	Backup path when Whisper or audio conversion fails	Resilient transcription layer
MediaPipe Face Mesh	Eye-contact approximation and face landmarks	Lightweight real-time facial landmark extraction	Visual monitoring
MediaPipe Pose	Posture analysis	Detects body posture cues during interview	Behavior tracking
MediaPipe Hands	Hand movement tracking	Captures gesture movement and fidgeting	Behavior tracking
YOLOv8	Object detection	Helps detect external objects or cheating scenarios	Cheating/suspicious-object monitoring
Judge0	Code compilation/execution	Enables runnable coding interviews inside the browser flow	Programming round

9. Feature Set and How It Works

- Resume-based interview question generation: extracts resume text and tailors questions to the candidate's background.
- Voice-based interview interaction: allows spoken answers and audio-driven interview behavior.
- Coding interview support: uses Monaco Editor for code input and Judge0 for execution/verification.
- AI answer evaluation: Gemini evaluates technical responses and produces scoring plus improvement notes.
- Speech transcription: Whisper converts audio answers to text before evaluation.
- Behavior monitoring: OpenCV + MediaPipe track webcam-based cues such as eye contact, posture, and gestures.
- Object detection and cheating detection: YOLOv8 identifies suspicious external objects in the frame.
- Evidence screenshot capturing: periodic snapshots provide supporting evidence for the report.
- Automated PDF reports: ReportLab / FPDF / PyMuPDF generate a polished output for later review.
- Cloud report storage: Supabase stores reports and returns secure signed URLs for download.

10. Architectural Changes and Improvements

- Added async monitoring using Python threading.
- Implemented fallback report generation when Gemini fails.
- Added a monitoring-status polling endpoint.
- Improved the PDF generation pipeline.
- Integrated YOLOv8 object detection support.
- Improved session handling architecture.
- Added evidence image capture.
- Improved backend logging and debugging visibility.
- Made report storage compatible with both local and cloud storage.
- Shifted from local PDF storage to Supabase cloud storage.
- Implemented signed URLs for secure report downloads.
- Replaced server-side TTS with browser SpeechSynthesis.
- Reduced Gunicorn workers to optimize memory usage.
- Introduced lazy-loading for Whisper models.
- Moved from hardcoded URLs to environment-based configuration.
- Improved AI response parsing and fallback handling.
- Added Docker support for reliable deployment.
- Separated frontend and backend into deployment-ready modules.
- Introduced Monaco Editor + Judge0 for coding rounds.
- Added Whisper AI for better offline speech recognition.
- Moved from simple evaluation to detailed Gemini AI analysis.
- Implemented Render deployment workflow for scalability.
- Added dynamic report generation with graphs and analytics.

11. Future Scope

- Real-time cheating detection
- Emotion analysis
- Multi-face detection
- Recruiter analytics dashboard
- JWT authentication
- Interview history tracking
- Performance graphs and analytics
- Live AI coaching during interviews
- Advanced recruiter-ready scoring system
- Upgrade browser TTS to Google Cloud TTS or ElevenLabs
- Implement authentication and user dashboards using Supabase Auth
- Add advanced analytics and candidate scoring dashboards
- Use larger Whisper/Gemini models on higher-memory infrastructure
- Add interview history and progress tracking
- Introduce live coding interview environments

- Integrate AI-driven resume improvement suggestions

12. Gallery

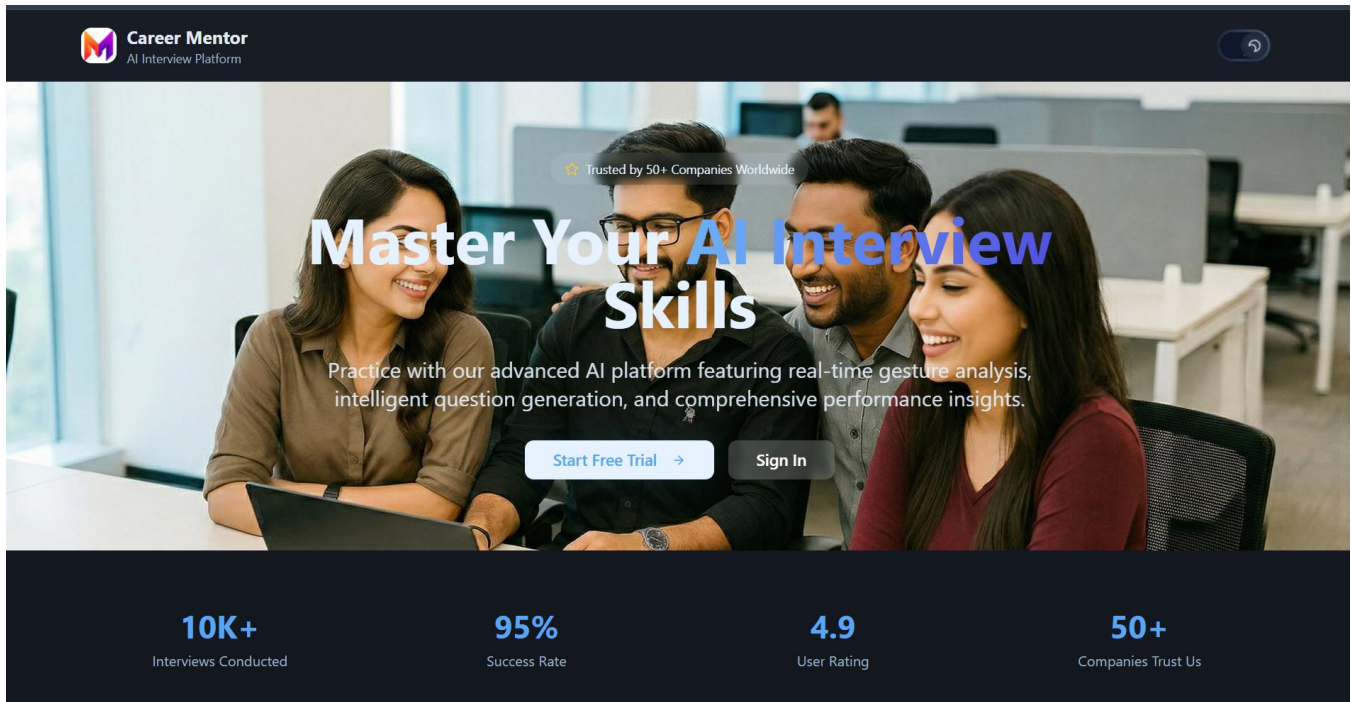


Figure 1: Home Page

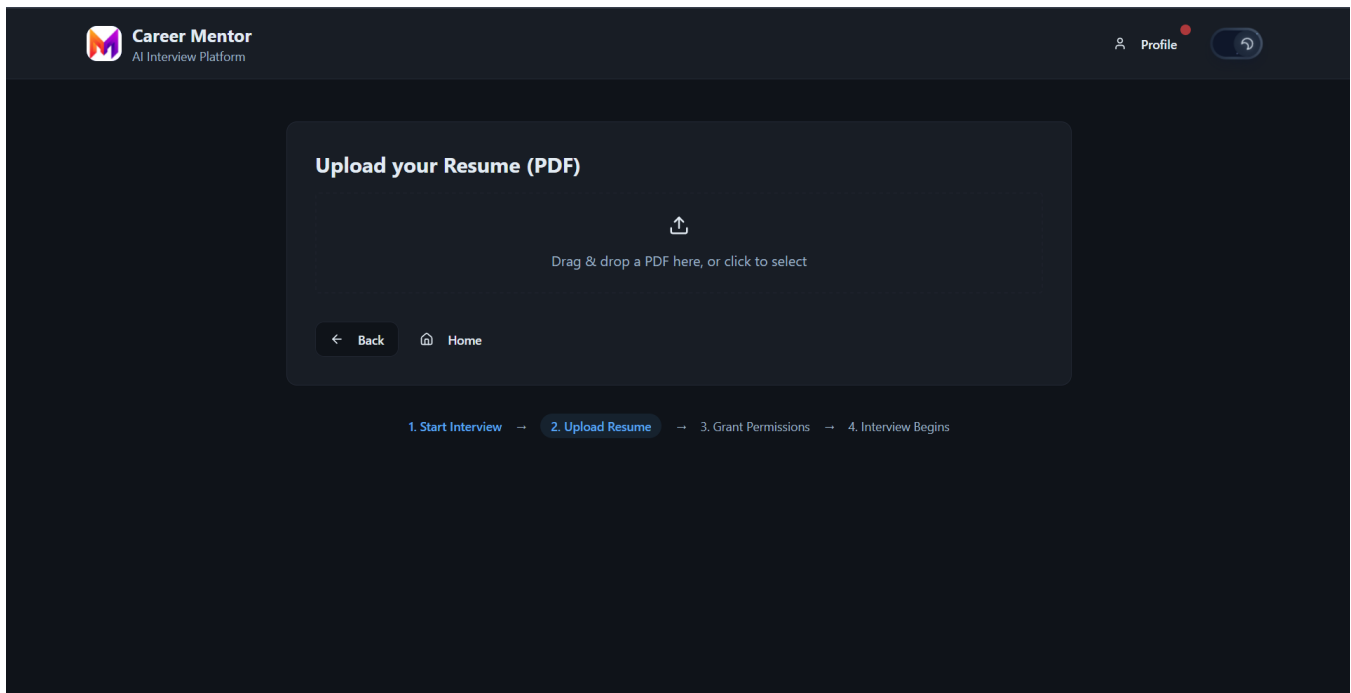


Figure 2: Upload your Resume/CV in PDF format

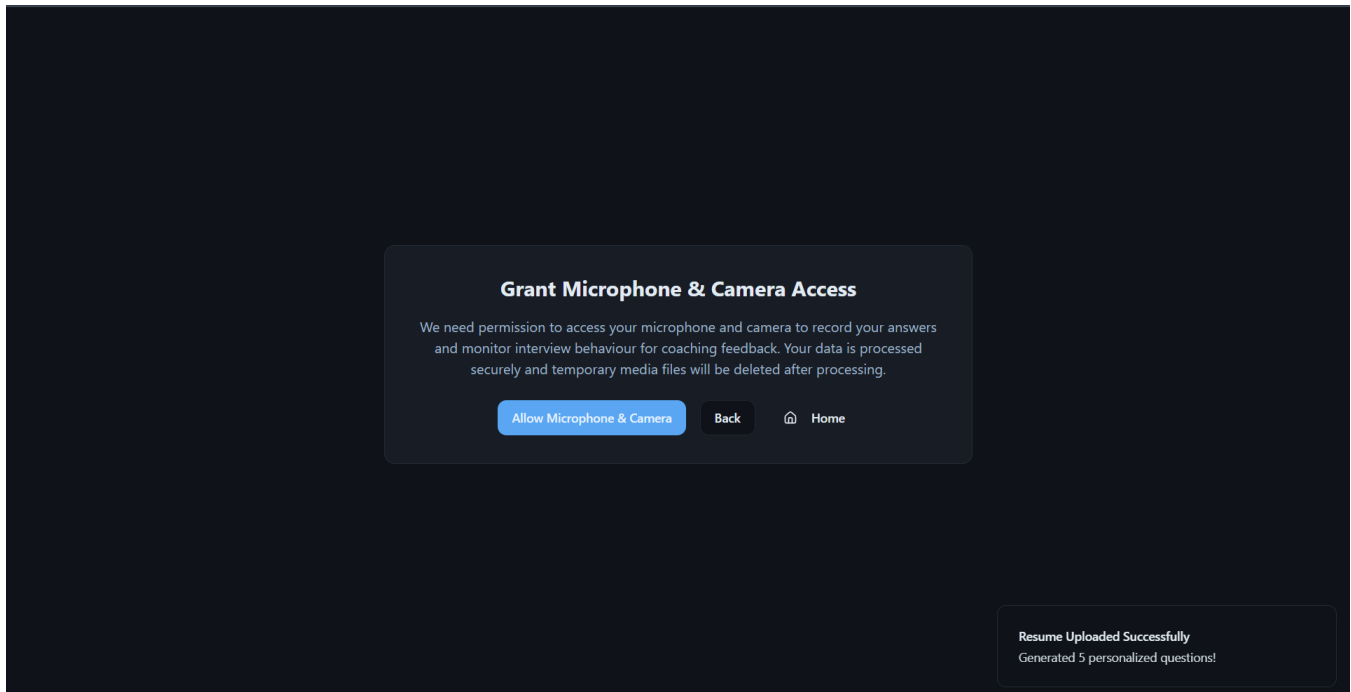


Figure 3: Allow Access to Camera and Microphone to capture your answers and activity.

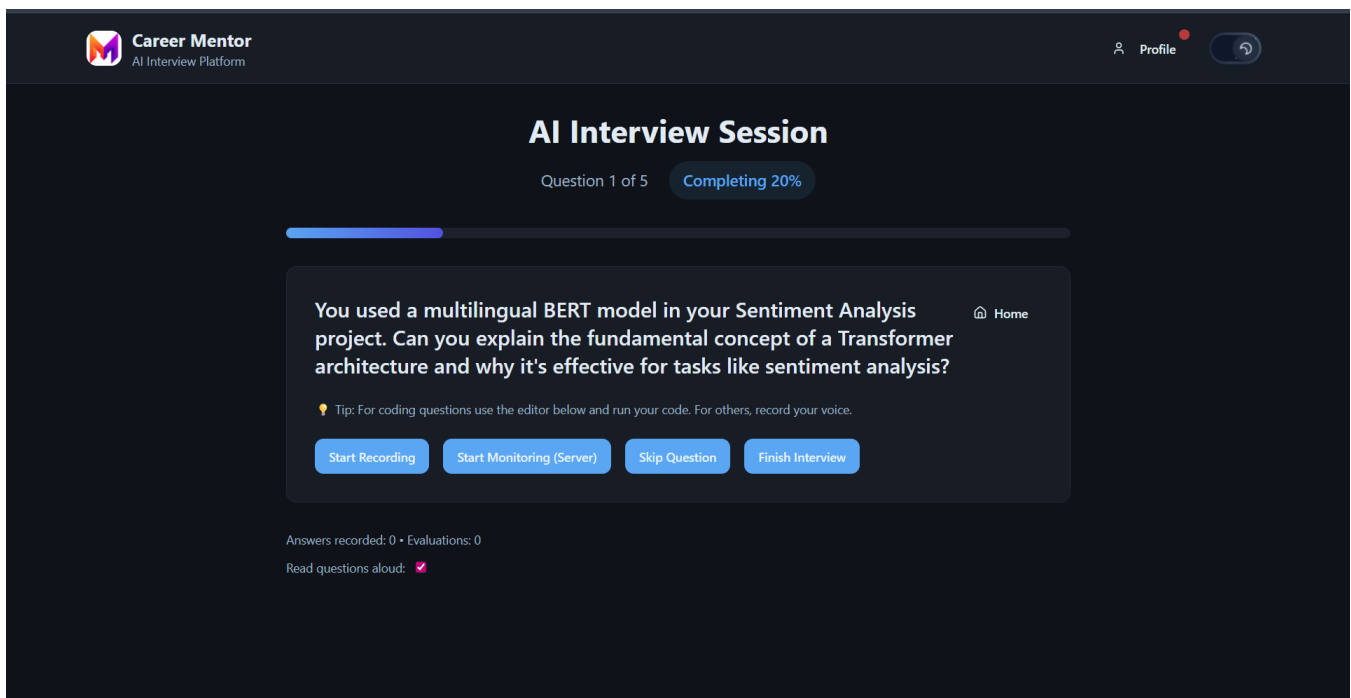


Figure 4: First question based on your Resume

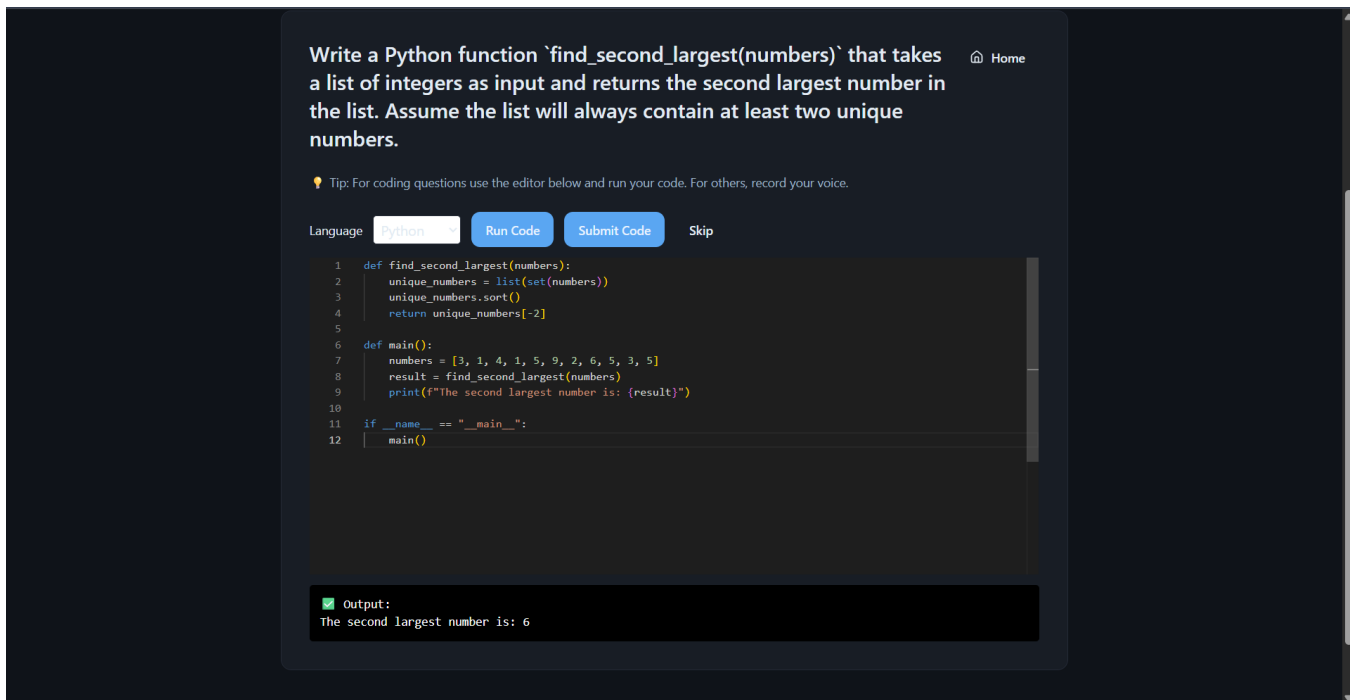


Figure 5: Coding question (Supported language: Python , C++ and Java

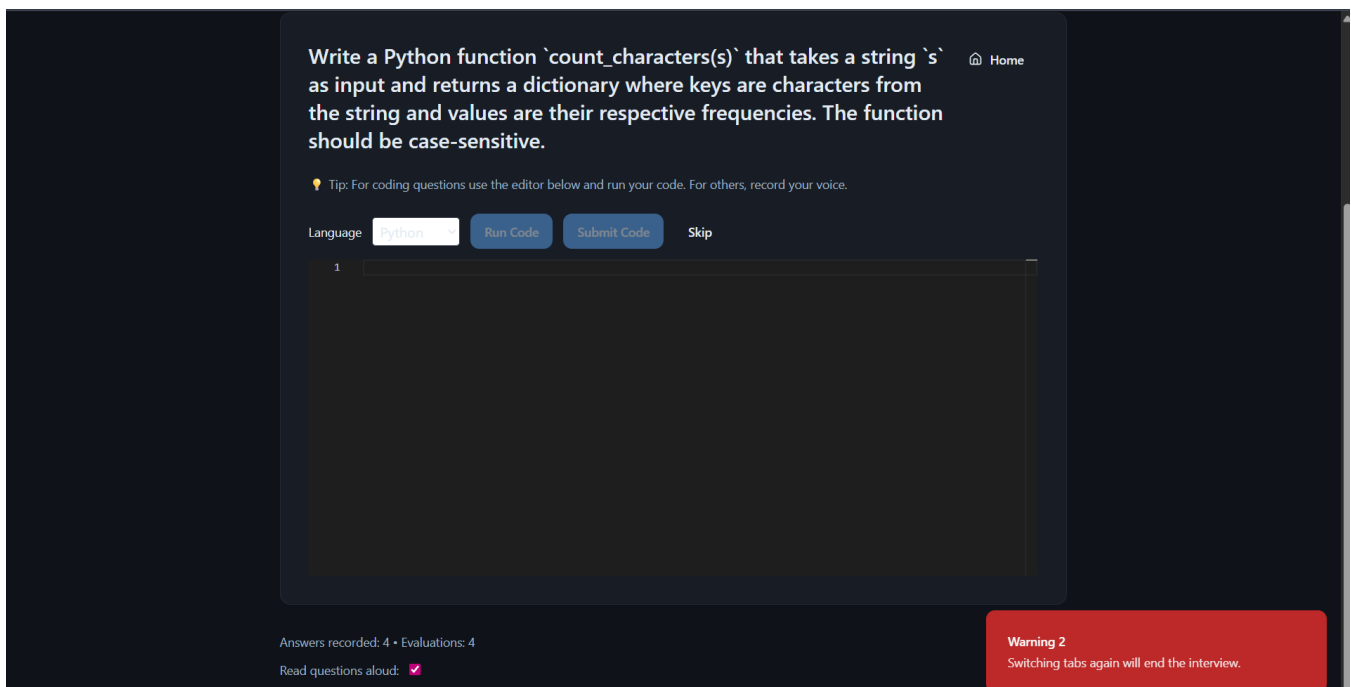


Figure 6: Switching Tab warning, Interview will be automatically submitted if 3rd time tab changed.

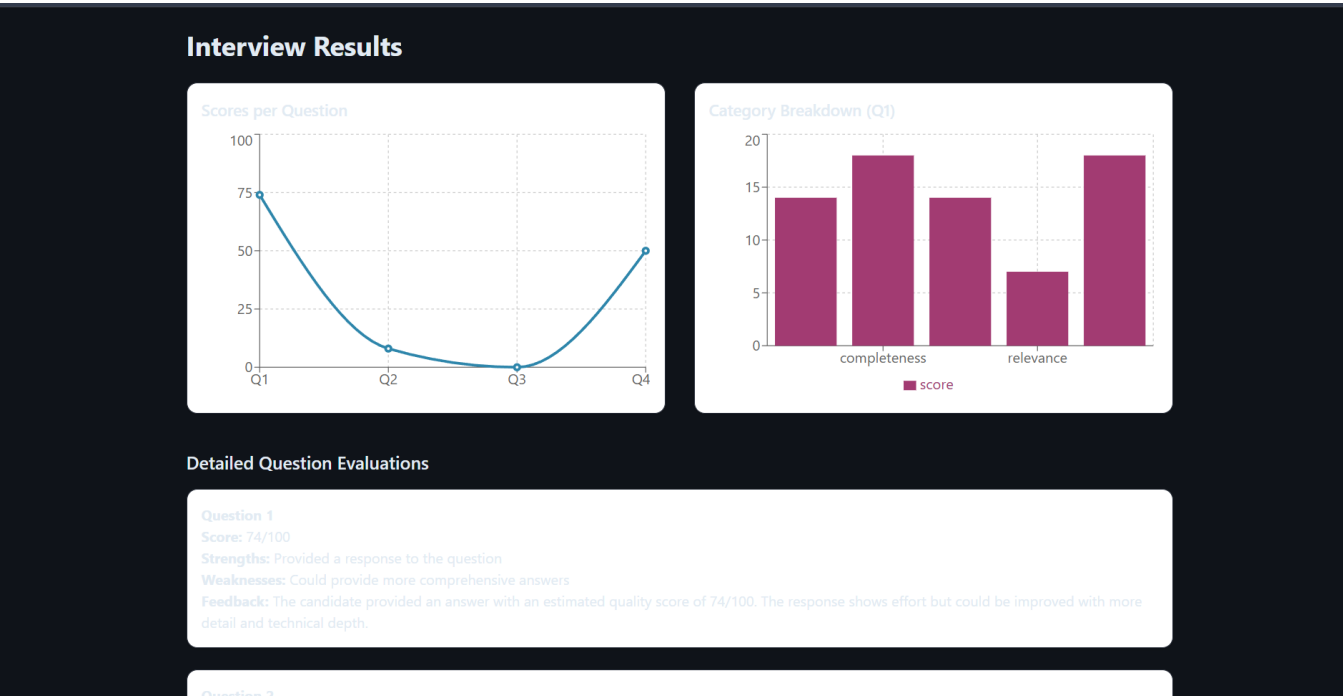


Figure 7: Analysis of your answers

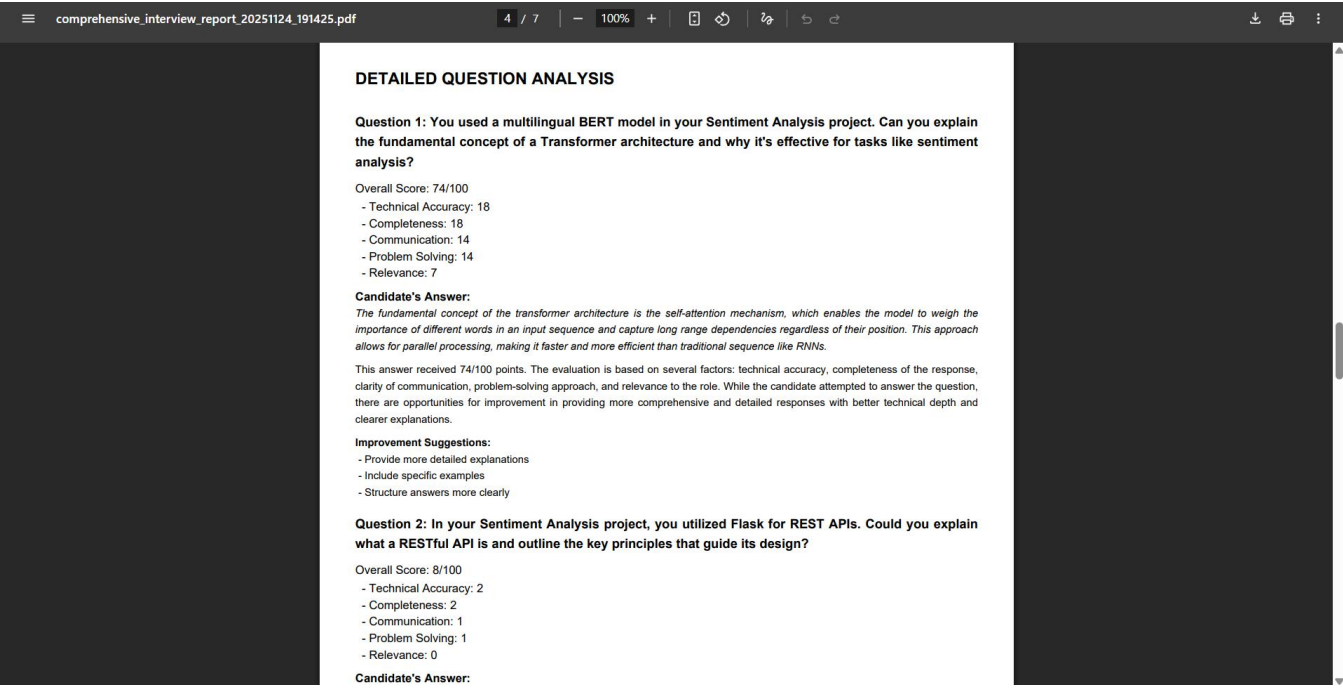


Figure 8: Detailed Report

13. Final Recruiter-Ready Summary

Career Mentor is no longer just an interview simulator. It is a full-scale AI-powered interview ecosystem that integrates Generative AI, speech recognition, coding evaluation, computer vision, cloud storage, analytics, and automated reporting.

From an engineering perspective, the project shows strong command over full-stack development, asynchronous backend design, real-time video analysis, API integration, deployment optimization, and production-style debugging. The final implementation is especially impressive because it combines user experience, AI reasoning, live monitoring, and reporting into one coherent workflow.

For a recruiter, the most valuable signals are clear: this project solves a real career-preparation problem, uses multiple AI modalities responsibly, handles deployment constraints, and demonstrates practical problem-solving under real system failures.